

بسم الله الرحمن الرحيم



GIT Elimination Disorders

By



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Human defecation

- **Human defecation involves integrated and coordinated sensorimotor functions which are orchestrated by central, spinal, peripheral (somatic and visceral) and enteric neural activities.**
- **Such activities act on a morphologically intact GIT (entailing the final common path, the pelvic floor, and anal sphincters).**

Sensorimotor functions

Central Neural Activities

Final Common Path

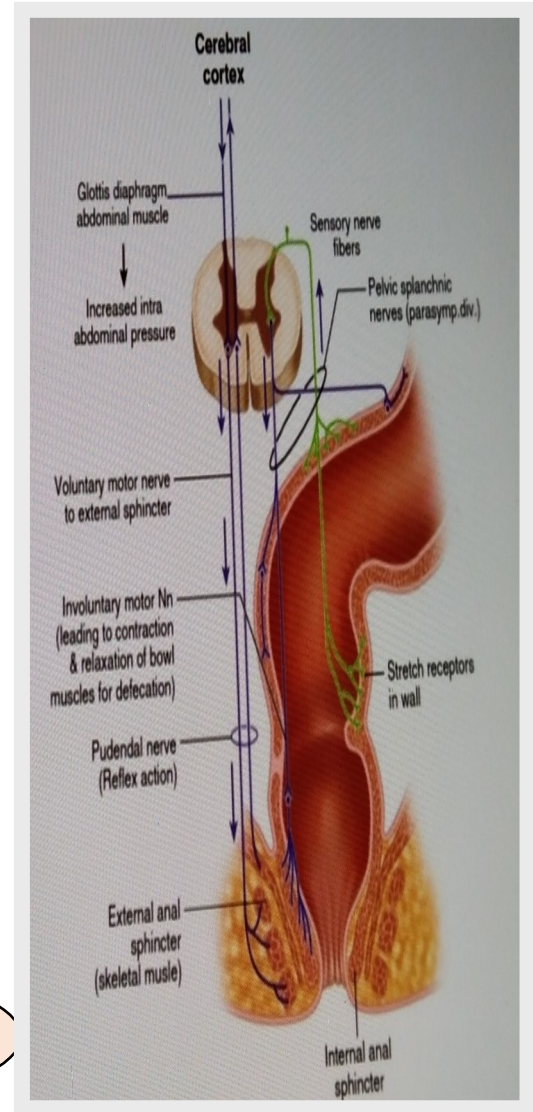
Spinal activities

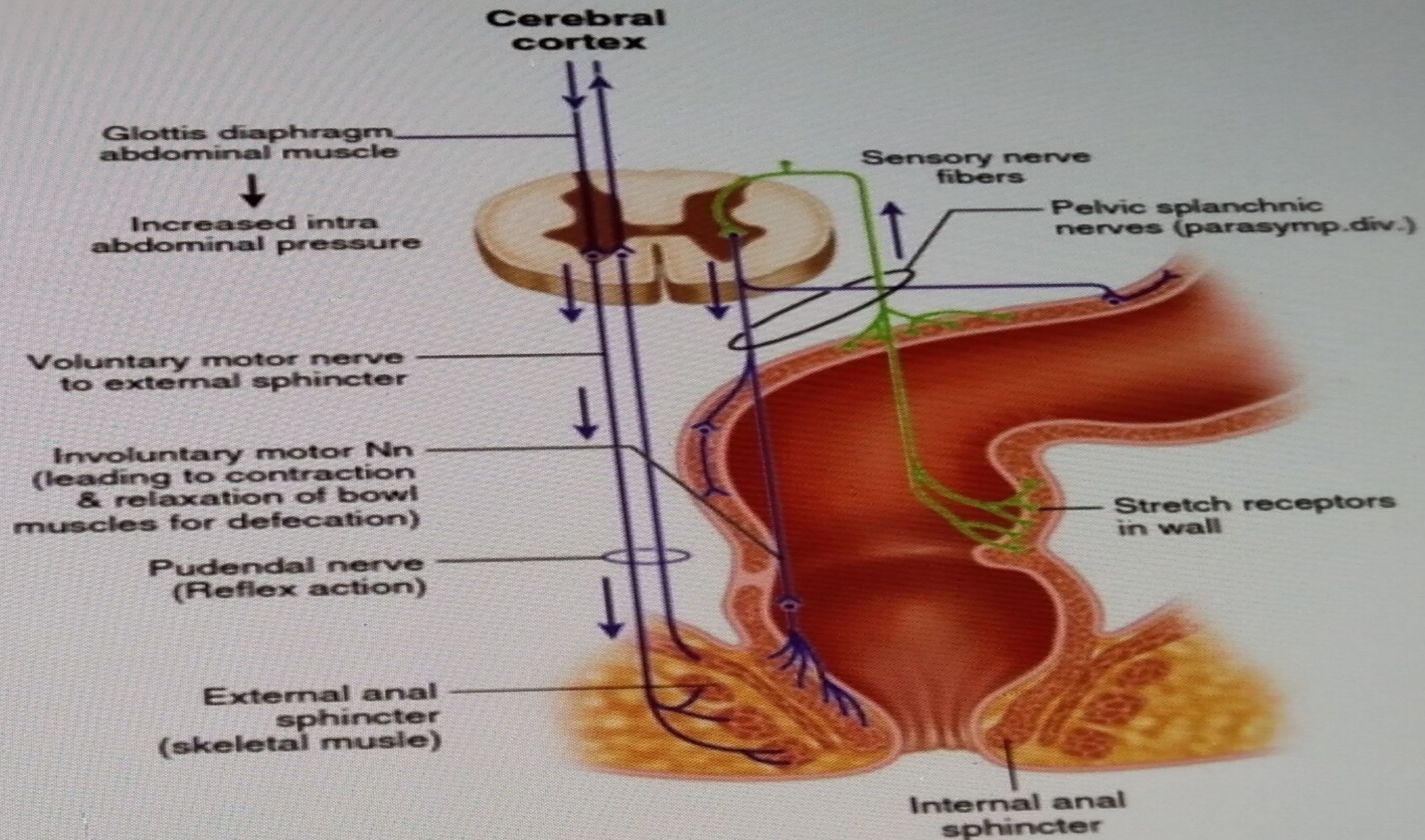
The Pelvic Floor

Peripheral Neural Activities

Enteric Neural Activities

Anal Sphincters





Encopresis (Paradoxical diarrhea)

- It is voluntary or involuntary passage of fecal matter outside of toilet trained contexts **(fecal soiling oftenly into their undergarments)** in children who are four years or older and after exclusion of organic causes.
- It is derived from the ancient Greek word (egkopresis)
- In adults, it is sometimes called fecal leakage (FL), fecal soiling, or fecal seepage)

Epidemiology

- **Age : in children 4 yrs or older (or equivalent developmental level)**
- **Gender: it is far common in males than females (6: 1)**
- **Prevalence: 1 - 3 %**
- **The onset is oftenly benign associated with toilet training with demands that the child sits for long time and intense negative parental reaction to feces**

Reversing of school enuresis is another

Epidemiology

- Feuding parents
- Feuding siblings
- Moving
- Parental discord or divorce

- Inhibit toilet behavior and promote constipation
- A precipitating cause may become less relevant as chronic stimuli predominate

Subtypes

Primary encopresis

- **At the age of 4 yrs or more**
- **1 - 3%**
- **More common in males (6:1)**

Secondary encopresis

- **Between the age of 7-8 yrs**
- **1.5%**
- **More common in males (3:1)**

Subtypes

80-95%

**With Constipation
and overflow
Incontinence**

- **Poorly formed feces**
- **Continuous leakage**
- **During sleep and waking hours**

5-20%

**Without Constipation
and overflow
incontinence**

- **Well formed feces**
- **Intermittent soiling**
- **During waking hours**
- **In a prominent place**
- **Comorbidity usually identified**

Comorbidity/Associations with secondary encopresis

- **Conduct disorder**
- **Oppositional defiant disorder**
- **Large anal insertions (repeated)**
- **Sexual and or physical abuse**
- **Chronic encopresis with desensitization of the rectum and anal canal**

- **Birth of a sibling**
- **Parental discord or divorce**
- **ID**

• **ASD/Other PMDs**

Pathogenesis of subtype I

- Constipation with reflexive stool withholding by

Physiological

Psychological/
Behavioral

- Delayed physical maturation
- Inappropriate toilet training

Neurological

Surgical/Medical

Pathophysiology

- **Normally, the colon removes excess water from feces**
- **If stool remains in the colon too long (due to conditioned withholding or incidental constipation), so much water is removed that the stool becomes hard and painful to expel in an ordinary bowel movement.**
- **A vicious circuit can develop**

Pathophysiology

**Avoidance of
moving bowel**

**Avoidance of expected
painful toilet episode**

This results in deep conditioning of the holding response which leads to RAIR or animus that continues to be there even under anesthesia or

Pathophysiology

- **Continuous building up of hard stools**
- **Stretching the colon or rectum to the degree that the normal feeling or sensation of the impending bowel movements does not occur**
- **Eventually, softer stool leaks around the blockage and cannot be withheld by the anus ending with soiling**

Pathophysiology

- **The child typically has no control over these leakage accidents**
- **May not be able to feel that they have occurred or about to occur due to the loss of sensation in the rectum and RAIR**
- **With resulting strong emotional reactions from failed and repeated attempts to control this highly aversive bodily product**
- **These emotional reactions may complicate conventional treatments with stool softeners, sitting demands, and behavioral strategies**

Diagnosis (DSM5 criteria)

- 1. Repeated passage of feces into inappropriate places (e.g. underwear or floor) whether voluntary or unintentional**
- 2. At least one such event a month for at least 3 months**
- 3. Chronological age of at least 4 years (or equivalental developmental level)**
- 4. The behavior is not exclusively due to a physiological effect of a substance (laxatives) or a general medical condition, except through a mechanism involving constipation**

Clinical manifestations

- **Involuntary or voluntary soiling of the underwear**
- **Those with constipation may experience**
 - **Anorexia**
 - **Abdominal pain**
 - **Pain on defecation**
 - **Less bowel movements**
 - **Have hard or soft stool**

Clinical manifestations

- **Age and gender**
- **Primary or secondary**
- **Timing (Day/Night)**
- **Location**
- **Bowel habits (frequency, consistency, size)**
- **Melena/hematochezia**
- **Pain with defecation**
- **Dietary habits**

Assessment

- **Abdominal tenderness/distension**
- **Height/weight**
- **Neurological assessment**
- **Skin evaluation**
- **Anorectal examination**

Investigations

- **Plain pelvi-abdominal X ray**
- **Pelvi-abdominal sonar**
- **Stool collection and analysis**
- **Barium enema/Rectal biopsy**
- **CBC, LFT, RFT**

Treatment of encopresis associated with constipation

- **Three pronged approach is recommended**

1. **Cleaning out (enemas, laxatives, or both)**

2. **Stool softeners use**

3. **Scheduling sitting times, typically after meals**

Treatment of encopresis associated with constipation

- Dietary program is recommended
 - 1. Reduction of intake of constipating food as dairy, peanuts, cooked carrots, and bananas
 - 2. Increase the intake of food rich with fibers as bran, whole wheat products, fruits, and vegetables
 - 3. Higher intake of water and liquids as juices (take care of tooth decay)
 - 4. Limit drinks with caffeine including cola drinks and tea
 - 5. Provide well balanced meals and snacks with limitation of fast/junk food s (high in fats and sugar)
 - 6. Limit whole milk to 500 ml/day for a child over 2 years of age

Treatment of encopresis associated with constipation

- 1. Family counseling and advices
(accessible, regular, non threatening
toilet training)**
- 2. CBT (Motivational system (contingency
management system))**
- 3. Group therapy/family therapy/support
groups**

Treatment of encopresis associated with constipation

- **Biofeedback**

- **Acupuncture**

- **Surgical intervention**

1. Cleaning out (enemas, laxatives, or both)

2. Stool softeners use

3. Scheduling sitting times, typically after meals

Thank You

